Fiscal Year:	FY 2025	**Task Last Updated:**	FY 02/03/2025
PI Name:	Seidler, Rachael D. Ph.D.		
Project Title:	Recovery Timeline of Spaceflight-Induced Central Nervous System Changes		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **EVA**:Risk of Mission Impacting Injury and Compromised Performance and Long-Term Health Effects due to EVA Operations (3) **SANS**:Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS) (4) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: PI moved to University of Florida in July 2017; previous affiliation was University of Michigan.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2019-2020 HERO 80JSC019N0001-HHCBPSR, OMNIBUS2: Human Health Countermeasures, Behavioral Performance, and Space Radiation-Appendix C; Omnibus2-Appendix D
Start Date:	03/30/2021	**End Date:**	08/31/2033
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 08/31/2033 per V. Lehman/JSC (Ed., 4/18/23). NOTE: End date changed to 03/31/2029 per L. Juliette/JSC (Ed., 5/3/22).		
Key Personnel Changes/Previous PI:	March 2022 Report: Babette Brumback, Ph.D., CoInvestigator, has retired from the University of Florida and is no longer with the project.		
COI Name (Institution):	Wood, Scott Ph.D. (NASA Johnson Space Center)		
Grant/Contract No.:	80NSSC21K0813		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Our group has reported an upward shift of the brain within the skull following spaceflight, which results in apparent reduced gray matter volume in inferior and frontal brain regions, and apparent increased volume in superior and posterior regions, as measured by magnetic resonance imaging (MRI). Another recent paper that we have published reports free water (fluid in the ventricles and extracellular space) changes in the brain with spaceflight, and degradation of sensory and motor white matter pathways (Pasternak O et al., "Spaceflight-Associated Brain White Matter Microstructural Changes and Intracranial Fluid Redistribution." JAMA Neurol. 2019 Apr 1;76(4):412-419. https:// . PMID: 30673793; PMCID: PMC6459132). Some of these measures show recovery to preflight levels by six months postflight, whereas others do not. For example, in two crewmembers who spent ~12 months in space, free water recovers only 75% by six months postflight. We have also observed increases in ventricular volume with spaceflight, ranging from 5 – 35% across astronauts. These changes exhibit little recovery by six months postflight, raising the possibility that these effects persist for prolonged durations. The brain's glial lymphatic (or 'glymphatic') pathway was identified in humans in just the past few years. It has been suggested that "ocular glymphatic" and cerebral lymphatic dysfunction may contribute to optic disc edema in astronauts, which is one sign of Spaceflight Associated Neuro-ocular Syndrome (SANS). SANS affects up to 50% of long-duration astronauts and poses significant health concerns. Many of these glymphatic vessels are found at the top of the brain and in the human extravascular visual system. Thus, the fluid and brain positional shifts that occur in microgravity may slow the rate of clearance of substances through this system. We are currently testing the impact of elevated CO2 on glymphatic clearance supported by an Office of Naval Research (ONR) grant (awarded to Kernagis and Seidler). The long-term health and functional consequences of these spaceflight-induced brain changes remain unknown, but our preliminary data suggest little recovery of fluid shifts that occur with one year in space when measured six months after return. Thus, we propose the following aims in response to Topic 3 of 80JSC019N0001-HHCBPSR: Aim 1) Determine the recovery timeline of spaceflight-induced ocular changes (e.g., signs of SANS such as optic disc edema) and changes in brain structure, function, and fluid shifts, measuring out to five years postflight. Aim 1a will comprise prospective assessments, while Aim 1b will leverage follow up testing on crewmembers who participated in our Neuromapping flight study and those crewmembers who have existing retrospective pre and postflight MRI scans. Aim 2) Examine longitudinal data for markers of long term health consequences and determine their association with persistent, spaceflight-induced ocular / brain changes. This project will elucidate the persistence of spaceflight-induced ocular and brain changes, as well as quantify their association with long-term health. Determining this is critically important as we extend the duration and distance of human space travel in the near future. Ed. Note (2/29/24): This project has been combined with the work of three other Principal Investigators (PIs) who responded to the same solicitation with independent proposals. The protocols and aims from all four PIs have been integrated into a revised proposal that was delivered to NASA Human Health Countermeasures (HHC) and Human Factors and Behavioral Performance (HFBP) Element Scientists in January 2022. For information on the four related investigations, see: • The Long-Term Consequences of Spaceflight on Brain and Eye Health (PI: Bershad). • Long-Term Brain Structural and Functional Consequences of Spaceflight (PI: Basner). • Investigating Long-term Structural and Functional Changes in the Eye and Brain After Spaceflight (PI: Macias). • Recovery Timeline of Spaceflight-Induced Central Nervous System Changes (PI: Seidler).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project has the potential to benefit life on Earth by leading to a greater understanding of central nervous system plasticity.
	INTRODUCTION Spaceflight-associated neuro-ocular syndrome (SANS) describes structural and functional alterations of the eye that develop in many astronauts during long-duration spaceflight missions (>6 months). Understanding the consequences of the structural and functional changes to the eye and brain that endure long after return from spaceflight is essential for future exploration-class spaceflight missions. PURPOSE The primary purpose of this project is to quantify long-term changes in ocular and brain structure that remain or develop long after spaceflight and to determine the functional consequences of those changes. Specifically, we will characterize the recovery and long-term structural and functional alterations to the eye and brain for 5 years following 6-month spaceflight missions. We will also characterize the structural and functional changes to the eye and brain, including cognitive changes, in crewmembers who have participated in past missions, calling them back several years after their most recent flights. METHODS Advanced techniques will be used to study the eyes and brain of astronauts. For one set of experiments, preflight, inflight, and 5 years of postflight results from astronauts (n = 10) who have participated in ≥6-month missions will be compared to those from age and sex-matched non-astronaut control subjects (n=13). In a second set of experiments, outcomes from astronauts who had previously participated in ≥6-month missions (n=30) will be compared to control subjects that do not have standard duration spaceflight experience (n=30). In addition to characterizing the effects observed for individual variables, all data collected during the study will be analyzed using an integrative approach to identify relationships across ocular measures, brain measures, individual crewmembers, and time. Ocular structure and function will be evaluated using MRI and 3D structural analysis, optical coherence tomography (OCT), angiography, electroretinography (ERG), visual field analysis, pneumotonometry (intraocular pressure), visual field analysis, optical biometry, and dynamic vessel analysis. In collaboration with the integrated project team, magnetic resonance imaging (MRI) will be used to assess brain structural changes, and functional MRI (fMRI) and a series of cognitive batteries will be used to evaluate brain function. To better understand the neurophysiological mechanisms underlying anticipated changes, biomarkers associated with specific structural and functional brain alterations will be measured in blood

Task Progress:	samples. To complement the research measures, medically relevant test results already collected by the Space Medicine Operations Division will be shared. PROGRESS DURING THIS REPORTING PERIOD This project was combined with the work of three other Principal Investigators (PIs) who responded to the same solicitation with independent proposals. The work was rescoped and a revised, integrated proposal was delivered to NASA Human Health Countermeasures (HHC) and NASA Human Factors and Behavioral Performance (HFBP) Element scientists in 2022. The project was approved as a NASA Virtual Specialized Center of Research in 2023. This Long-Term Health (LTH) Virtual NASA Specialized Center of Research (VNSCOR) has a ground and flight component, and therefore two Research Operations and Integration (ROI) coordination teams. Dr. Macias' PI team is supporting bi-weekly meetings with ROI to implement both portions of the projects. In addition, all Co-PI teams meet bi-weekly with Dr. Macias' PI team to coordinate the overall implementation effort of the ground and flight components of the project. The "LTH Call Back" retrospective arm will focus on investigating the structural and functional changes in the eye, brain, and cognition in astronauts who previously participated in long-duration missions, calling them back at least 5 years after their previous flights to determine changes that may endure long after return from spaceflight. Major accomplishments during this reporting period for this sub-study include: • The Lifetime Surveillance of Astronaut Health (LSAH) data share request was approved by the LSAH Advisory Board. • An LSAH epidemiologist was added to the study team to help with data analysis. • The MRI facility contract was finalized, and two MRI dry runs were completed at Houston Physicians' Hospital. • One internal dry run was completed in the NASA Cardiovascular and Vision Laboratory. • The neuropsychological and neurological Co-PI team completed a dry run at NASA Johnson Space Center (JSC). • The Crew Office approved the Informed Consent Briefing (ICB). • Recruitment efforts were initiated by ROI and are ongoing. • The Test Readiness Review (TRR) package was developed and is undergoing internal review. Ongoing forward work with ROI for "LTH Call Back" includes: • Approval of the TRR package (planned on December 2024). • Initiate recruitment and testing of astronaut and matched control participants. The "Brain Health" component is prospective and will investigate the development and recovery of structural and functional changes in the eye, brain, and cognition in astronauts participating in standard or extended duration missions and will follow them out 5 years postflight. Major accomplishments for the Flight Study during this reporting period include: • ROI submitted the Feasibility Assessment to the HHC and HFBP Elements and the sub-study was approved for Select for Flight in December 2024. • The LSAH data share request was approved by the LSAH Advisory Board. • An LSAH epidemiologist was added to the study team to help with data analysis. Ongoing future work with ROI for "Brain Health" includes (pending select for flight determination by the NASA Human Research Program/HRP): • Completion of two MRI dry runs for the Brain Health MRI scanning protocol. • Development and approval of a TRR package (targeting January 2025). • Development and approval of ICB presentation. • Pitch ICB starting with Crew-12. Presentation: K.J. Carter, M. Basner, E. M. Bershad, R. Seidler, A. C. Stahn, S. S. Laurie, B. R. Macias. Investigating long-term structural and functional changes to the eye and brain after spaceflight. Human Research Program Investigators' Workshop, Galveston, TX, February 13-16, 2024.
Bibliography Type:	Description: (Last Updated: 03/12/2026)
Abstracts for Journals and Proceedings	Carter KJ, Basner M, Bershad EM, Seidler RD, Stahn AC, Laurie SS, Macias BR. "Investigating long-term structural and functional changes to the eye and brain after spaceflight." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025
Articles in Peer-reviewed Journals	Seidler RD, Mao XW, Tays GD, Wang T, Zu Eulenburg P. "Effects of spaceflight on the brain." Lancet Neurol. 2024 Aug;23(8):826-35. Review. Epub 2024 Jun 27. http://dx.doi.org/10.1016/S1474-4422(24)00224-2 ; PMID: 38945144 , Aug-2024